

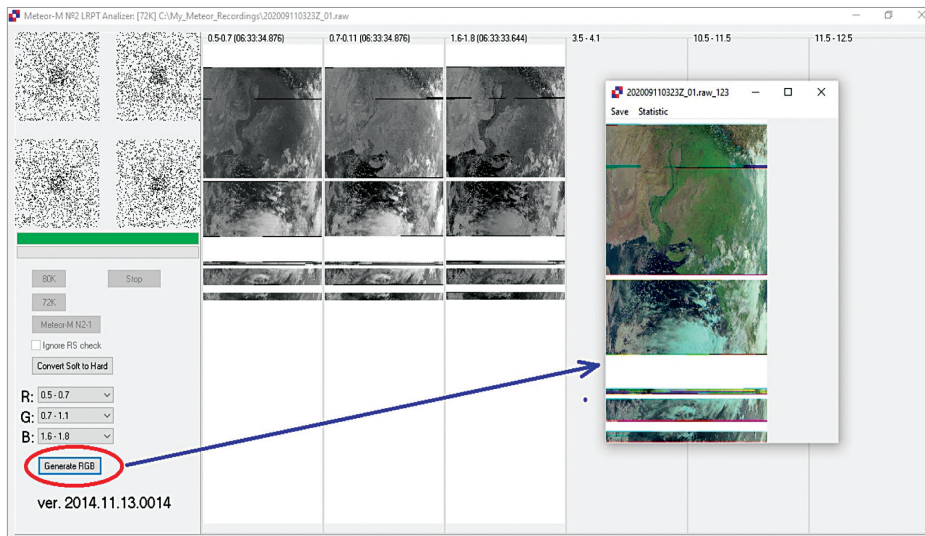


Fig. 9(a): Image extraction from LRPT Analyser software with three visible channels and the generated image



Fig. 9(b): Seasonal change in the Meteor M2 spectral bands with five channels

erly, the Lock Detector (red dot in Fig. 8(a)) turns green and also the Constellation Diagram towards the right starts showing the four phase-amplitude dots of the QPSK signal, as illustrated in Fig. 8(b).

As the demodulation progresses, the status/progress bar also slides towards right. The only problem with this software is that it does not stop automatically, even after all the file data has been processed. You need to press the Stop button to stop it and then quit the program.

This completes the manual demodulation processes. Next, we shall see the final step that is ap-

plicable to both the approaches, that is, to extract the actual image file(s) from this demodulated data.

Image extraction, the final step

This is the final step in our effort to demodulate the earth imagery from the Meteor satellite data. From the above discussions, either through the plug-in or through the manual demodulation, we get the .S or the .RAW file. At this stage, we shall require the Off-line Meteor LRPT Analyser software, as mentioned earlier in this article.

Click on the 72k button in LRPT Analyser/Decoder software and

the file selection dialogue box opens. Choose either S file or RAW file. The software starts and shows the constellation diagram. If it finds the data properly, you shall see the grey-scale images displayed from the three active bands as sent by the Meteor satellite. Out of these three, there will be two in the visible/near-IR region. The third channel is switched depending upon the season to monitor the snow in the higher latitude countries during winter-time.

Once the decryption process is completed, you can click on the Generate RGB button and the image will be produced. Refer Fig. 9(a) and Fig. 9(b).

Conclusion

Images from each of these satellites, the NOAA satellites with Analogue APT weather imagery and the high resolution images from the Meteor M2, are quite useful for understanding the weather, cloud patterns, and cyclones. The author has also been experimenting with various antennas, impedance matching, and the antenna pre-amplifier for making this reception better in the best possible way and within acceptable

budgets! For a successful reception, a proper antenna and good cabling system is equally important. **EFY**

Prasanna Waichal holds a master's degree in electronics and has more than 25 years of work experience in industrial R&D for design and development of specialised and strategic electronics systems and academic teaching and research at post-graduate levels, both in India and abroad. His passions include radio reception and studying solar activities and their effect on ionosphere on earth. He is also a licensed amateur radio operator with call-sign VU3OOL. He claims to have discovered a method to predict earthquakes using very-low-frequency (VLF) multipath-multi-frequency propagation technique

